

Omar A. Khan

My research interests span **embodiment**, **interaction**, and **perception** in **XR (AR, VR, and MR)** environments, with a focus on avatars, multisensory perception, body perception, haptics, and tangible user interfaces. I seek to broaden our understanding of how next-generation computing interfaces shape human perception and behavior.

Education

- 2025 – Now **Ph.D. in Computer Science**, Drexel University, PA, USA.
Advisor: Tiffany D. Do
- 2021 – 2025 **B.Sc. in Computer Science**, University of Calgary, AB, Canada.
Advisor: Kangsoo Kim

Publications

Google Scholar: <https://scholar.google.com/citations?user=aVs159UAAAAJ>

Peer-Reviewed Journal Articles

- 2026 J1 **Omar A. Khan**, Bao Han Trinh, Lee Lisle, and Tiffany D. Do. "Having Dog Ears 'for Real': Effects of TVCG Active and Passive Haptics on Embodying Non-Human Body Parts in VR." To appear in *IEEE Transactions on Visualization and Computer Graphics*, 2026. doi:10.48550/arXiv.2606.26364
Acceptance rate: **16%**
- 2026 J2 **Omar A. Khan**, Junyeong Kum, Hyeongil Nam, Myungho Lee, and Kangsoo Kim. "Influence of Avatar TVCG Appearance and Target Distance on Locomotion Method Selection in Virtual Reality." *IEEE Transactions on Visualization and Computer Graphics*, 32(5):4668–4677, 2026.
doi:10.1109/TVCG.2026.3679092
Acceptance rate: **20%**
- 2025 J3 **Omar Khan**, Hyeongil Nam, and Kangsoo Kim. "Impact of Avatar-Locomotion Congruence on User TVCG Experience and Identification in Virtual Reality." *IEEE Transactions on Visualization and Computer Graphics*, 31(11):9878–9888, 2025. doi:10.1109/TVCG.2025.3616836
Acceptance rate: **8%**

🏆 **Honorable Mention Award at IEEE ISMAR 2025 (Top 3%)**

Peer-Reviewed Conference Proceedings

- 2026 C1 Nathalia Gomez, Haig Shamlan, **Omar A. Khan**, and Tiffany D. Do. "Listening and Mirroring: The VRST Effects of Verbal Attunement and Behavioral Mimicry on Social and Empathic Perceptions of Embodied AI Agents in VR." To appear in *Proceedings of 2026 ACM Virtual Reality Software and Technology*, pp. 1-10, 2026. doi:10.48550/arXiv.2609.27246
Acceptance rate: **27%**

Peer-Reviewed Extended Abstracts, Posters, and Workshop Papers

- 2026 E1 **Omar A. Khan**, Lee Lisle, and Tiffany D. Do. "Non-Human Body Parts in Social VR: A Survey of Furry SUI and Kemonomimi Communities." To appear in *Proceedings of the ACM Symposium on Spatial User Interaction*, 2026. doi:10.1145/3822518.3839315
Acceptance rate: **TBA**
- 2026 E2 Zaid Ahmed, **Omar A. Khan**, Hyeongil Nam, and Kangsoo Kim. "Exploring Experiential Differences CHI EA Between Virtual and Physical Memory-Linked Objects in Extended Reality." In *Proceedings of the ACM CHI Conference Extended Abstracts on Human Factors in Computing Systems*, no. 269, pp. 1–5, 2026. doi:10.1145/3772363.3798977
Acceptance rate: **38%**

- 2025 VRW E3 **Omar Khan**, Zaid Ahmed, Hyeongil Nam, and Kangsoo Kim. "TangibleMoments: Embedding XR Memories onto Physical Objects." In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pp. 1142–1146, 2025. doi:10.1109/VRW66409.2025.00227
- 2025 VRW E4 **Omar Khan**, Anh Nguyen, Hyeongil Nam, and Kangsoo Kim. "Investigating Visual Guide Cues in VR: Impacts of Virtual Humans and Symbol-Based Navigation on Real-World Performance and Experience." In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pp. 504–509, 2025. doi:10.1109/VRW66409.2025.00110
- 2025 VRW E5 **Omar Khan**, Hyeongil Nam, and Kangsoo Kim. "I look like a gorilla, but don't move like one!": Impact of Avatar-Locomotion Congruence in Virtual Reality." In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pp. 1202–1203, 2025. doi:10.1109/VRW66409.2025.00247
- 2024 VRW E6 **Omar Khan**, Anh Nguyen, Michael Francis, and Kangsoo Kim. "Exploring the Impact of Virtual Human and Symbol-Based Guide Cues in Immersive VR on Real-World Navigation Experience." In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pp. 883–884, 2025. doi:10.1109/VRW62533.2024.00238

Academic Service and Outreach

Peer Review

- 2027 ACM Conference on Human Factors in Computing Systems (CHI)
- 2027 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)
- 2026 Springer Virtual Reality
- 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2025 – 2026 ACM Symposium on Virtual Reality Software and Technology (VRST)

Student Volunteer

- 2025 – 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

Outreach

- 2026 – 2027 Event Coordinator, Drexel SCIS Doctoral Student Association

Awards and Honors

- 2025 🏆 **Best Paper Award Honorable Mention (Top 3%), IEEE ISMAR**
- 2025 **IDEAS Travel Fund**, University of Calgary – \$1,352
- 2024 **Undergraduate Research Symposium Faculty of Science Award**, University of Calgary – \$1,000
- 2024 **Summer Research Studentship Award**, Alberta Innovates – \$8,500
- 2024 **Graeme Bell Travel Award**, University of Calgary – \$1,350
- 2023 **Undergraduate Student Research Award**, NSERC – \$7,500
- 2021 – 2025 **Jason Lang and Alexander Rutherford Scholarships**, Government of Alberta – \$5,500 total

Mentoring Experience

- 2026 **Arijit Chakma** (Undergraduate Student, Drexel University)
- 2025 – 2026 **Bao Han Trinh** (Undergraduate Student, Drexel University)

Invited Talks

- 2026 **"Modern Research in Virtual Embodiment"**, guest lecture for CSE 566, Stony Brook University.